

YEAR 7 MATHEMATICS

Chapter 4

Fractions and percentages

WORKSHEETS

4

How to use this booklet

- 1** There is **one worksheet for each lesson**. It has three columns that get harder from left to right: **Easy**, then **Medium**, then **Challenging**.

- 2** Read the **speech bubble** at the top of each column: it tells you what to do. A new bubble means a new kind of question.

WE DO **Top of each column.** Your teacher works the examples on the board. **Copy every step** into the space.

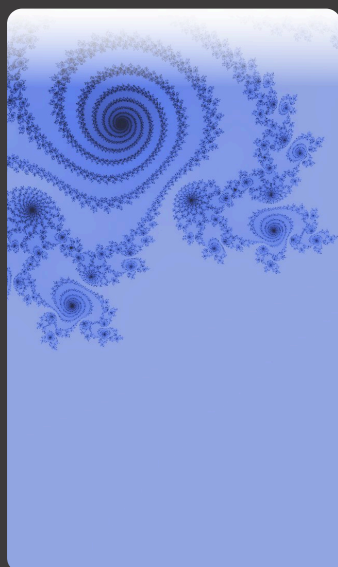
YOU DO **The rest of the column.** Your turn. Column 1: answer on the line. Columns 2 and 3: show your working on the lines.

Contents and progress tracker

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In this chapter you will

- write equivalent fractions and simplify
- order fractions
- add, subtract, multiply and divide fractions and mixed numerals
- find a fraction or percentage of a quantity
- solve fraction problems
- convert between fractions, decimals and percentages



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is a triple spiral on the edge of the top bulb. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

Easy

Questions 1–25

Simplify each fraction: divide the top and bottom by the same number.

WE DO With your teacher: copy the working.

E1 $\frac{4}{8}$ _____

E2 $\frac{6}{9}$ _____

YOU DO Your turn.

1 $\frac{35}{50}$ $\frac{7}{10}$

2 $\frac{3}{6}$ _____

3 $\frac{27}{30}$ _____

4 $\frac{10}{15}$ _____

5 $\frac{6}{12}$ _____

6 $\frac{4}{10}$ _____

7 $\frac{24}{30}$ _____

8 $\frac{12}{16}$ _____

9 $\frac{33}{36}$ _____

10 $\frac{3}{18}$ _____

11 $\frac{2}{6}$ _____

12 $\frac{28}{32}$ _____

Change each mixed numeral to an improper fraction.

WE DO With your teacher: copy the working.

E3 $2\frac{3}{4}$ _____

YOU DO Your turn.

13 $4\frac{2}{5}$ $\frac{22}{5}$

14 $5\frac{2}{3}$ _____

15 $4\frac{1}{2}$ _____

16 $4\frac{1}{3}$ _____

17 $5\frac{3}{5}$ _____

18 $2\frac{1}{6}$ _____

19 $4\frac{9}{10}$ _____

20 $2\frac{1}{2}$ _____

21 $3\frac{1}{3}$ _____

22 $2\frac{2}{3}$ _____

23 $6\frac{3}{4}$ _____

24 $3\frac{1}{5}$ _____

25 $3\frac{1}{4}$ _____

Medium

Questions 26–45

Find the missing number. What was the bottom (or top) multiplied by?

WE DO With your teacher: copy the working.

E1 $\frac{2}{3} = \frac{?}{9}$ _____

E2 $\frac{1}{2} = \frac{?}{16}$ _____

YOU DO Your turn.

26 $\frac{2}{5} = \frac{?}{15}$ $5 \times 3 = 15$

27 $\frac{3}{4} = \frac{?}{32}$ $4 \times 8 = 32$

$2 \times 3 = 6$ _____

$4 \times 8 = 32$ _____

$2 \times 3 = 6$ _____

$4 \times 8 = 32$ _____

28 $\frac{1}{4} = \frac{?}{24}$ _____

29 $\frac{5}{6} = \frac{?}{30}$ _____

$1 \times 6 = 6$ _____

$5 \times 6 = 30$ _____

$1 \times 6 = 6$ _____

$5 \times 6 = 30$ _____

30 $\frac{3}{8} = \frac{?}{24}$ _____

31 $\frac{2}{3} = \frac{?}{21}$ _____

$3 \times 8 = 24$ _____

$2 \times 10.5 = 21$ _____

32 $\frac{11}{12} = \frac{?}{96}$ _____

33 $\frac{1}{2} = \frac{?}{10}$ _____

$11 \times 8 = 88$ _____

$1 \times 10 = 10$ _____

34 $\frac{2}{5} = \frac{?}{25}$ _____

35 $\frac{3}{4} = \frac{?}{8}$ _____

$2 \times 12.5 = 25$ _____

$3 \times 2 = 6$ _____

36 $\frac{7}{8} = \frac{?}{48}$ _____

37 $\frac{1}{2} = \frac{?}{18}$ _____

$7 \times 6 = 42$ _____

$1 \times 9 = 9$ _____

38 $\frac{5}{6} = \frac{?}{54}$ _____

39 $\frac{1}{3} = \frac{?}{9}$ _____

$5 \times 10.8 = 54$ _____

$1 \times 9 = 9$ _____

40 $\frac{3}{5} = \frac{?}{40}$ _____

41 $\frac{1}{10} = \frac{?}{40}$ _____

$3 \times 13.33 = 40$ _____

$1 \times 40 = 40$ _____

42 $\frac{2}{3} = \frac{?}{6}$ _____

43 $\frac{7}{10} = \frac{?}{20}$ _____

$2 \times 3 = 6$ _____

$7 \times 2.857 = 20$ _____

44 $\frac{1}{5} = \frac{?}{20}$ _____

45 $\frac{1}{3} = \frac{?}{18}$ _____

$1 \times 20 = 20$ _____

$1 \times 18 = 18$ _____

46 $\frac{2}{3} = \frac{?}{9}$ _____

47 $\frac{12}{96}$ _____

$2 \times 4.5 = 9$ _____

$12 \div 8 = 1.5$ _____

48 $\frac{11}{12} = \frac{?}{96}$ _____

49 $\frac{24}{54}$ _____

$11 \times 8.727 = 96$ _____

$24 \div 2.25 = 10.666$ _____

50 $\frac{56}{104}$ _____

51 $\frac{24}{80}$ _____

$56 \div 1.857 = 30.181$ _____

$24 \div 3.333 = 7.2$ _____

Challenging

Questions 46–51

Simplify fully. Find the highest common factor (HCF) first.

WE DO With your teacher: copy the working.

E1 $\frac{12}{20}$ _____

E2 $\frac{36}{42}$ _____

YOU DO Your turn.

46 $\frac{32}{36}$

HCF of 32 and 36 is 4

$32 \div 4 = 8, 36 \div 4 = 9$

$= \frac{8}{9}$

47 $\frac{12}{96}$

HCF of 12 and 96 is 12

$12 \div 12 = 1, 96 \div 12 = 8$

$= \frac{1}{8}$

48 $\frac{24}{44}$ _____

49 $\frac{24}{54}$ _____

50 $\frac{56}{104}$ _____

51 $\frac{24}{80}$ _____

Easy

Questions 1–29

Which fraction is larger? The bottoms are the same, so compare the tops.



WE DO With your teacher: copy the working.

E1 $\frac{5}{12}$ or $\frac{3}{12}$

E2 $\frac{2}{10}$ or $\frac{9}{10}$

YOU DO Your turn.

1 $\frac{5}{8}$ or $\frac{3}{8}$

2 $\frac{1}{4}$ or $\frac{3}{4}$

3 $\frac{3}{4}$ or $\frac{2}{4}$

4 $\frac{4}{8}$ or $\frac{7}{8}$

5 $\frac{6}{9}$ or $\frac{8}{9}$

6 $\frac{8}{9}$ or $\frac{6}{9}$

7 $\frac{2}{5}$ or $\frac{3}{5}$

8 $\frac{7}{10}$ or $\frac{6}{10}$

9 $\frac{7}{10}$ or $\frac{3}{10}$

10 $\frac{1}{8}$ or $\frac{5}{8}$

11 $\frac{5}{11}$ or $\frac{7}{11}$

12 $\frac{4}{12}$ or $\frac{9}{12}$

13 $\frac{3}{9}$ or $\frac{4}{9}$

14 $\frac{1}{3}$ or $\frac{2}{3}$

15 $\frac{8}{9}$ or $\frac{1}{9}$

16 $\frac{2}{3}$ or $\frac{1}{3}$

17 $\frac{2}{7}$ or $\frac{1}{7}$

18 $\frac{4}{5}$ or $\frac{2}{5}$

19 $\frac{5}{10}$ or $\frac{6}{10}$

20 $\frac{4}{7}$ or $\frac{6}{7}$

21 $\frac{1}{5}$ or $\frac{2}{5}$

22 $\frac{3}{8}$ or $\frac{4}{8}$

23 $\frac{4}{12}$ or $\frac{8}{12}$

24 $\frac{1}{5}$ or $\frac{4}{5}$

25 $\frac{8}{12}$ or $\frac{2}{12}$

26 $\frac{1}{9}$ or $\frac{7}{9}$

27 $\frac{2}{9}$ or $\frac{7}{9}$

28 $\frac{2}{12}$ or $\frac{7}{12}$

29 $\frac{4}{7}$ or $\frac{2}{7}$

Medium

Questions 30–45

Which is larger? Write both with a common denominator first.



WE DO With your teacher: copy the working.

E1 $\frac{1}{3}$ or $\frac{7}{8}$

E2 $\frac{1}{10}$ or $\frac{7}{8}$

YOU DO Your turn.

30 $\frac{7}{10}$ or $\frac{3}{8}$

31 $\frac{7}{8}$ or $\frac{1}{4}$

$\frac{28}{40}$ and $\frac{15}{40}$

$\frac{7}{8}$ and $\frac{2}{8}$

larger: $\frac{7}{10}$

larger: $\frac{7}{8}$

32 $\frac{1}{2}$ or $\frac{1}{4}$

33 $\frac{1}{2}$ or $\frac{3}{10}$

$\frac{2}{4}$ and $\frac{1}{4}$

larger: $\frac{1}{2}$

34 $\frac{1}{4}$ or $\frac{4}{5}$

35 $\frac{9}{10}$ or $\frac{2}{3}$

36 $\frac{4}{5}$ or $\frac{2}{3}$

37 $\frac{1}{4}$ or $\frac{1}{5}$

38 $\frac{3}{4}$ or $\frac{1}{2}$

39 $\frac{1}{2}$ or $\frac{9}{10}$

40 $\frac{3}{5}$ or $\frac{3}{4}$

41 $\frac{1}{6}$ or $\frac{1}{3}$

42 $\frac{1}{2}$ or $\frac{2}{5}$

43 $\frac{1}{3}$ or $\frac{9}{10}$

44 $\frac{1}{2}$ or $\frac{3}{4}$

45 $\frac{1}{8}$ or $\frac{1}{2}$

Challenging

Questions 46–52

Write in ascending order (smallest first). Use a common denominator.



WE DO With your teacher: copy the working.

E1 $\frac{1}{12}$, $\frac{1}{4}$, $\frac{3}{4}$

E2 $\frac{2}{3}$, $\frac{1}{6}$, $\frac{5}{12}$

YOU DO Your turn.

46 $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{3}$

$\frac{6}{12}$, $\frac{9}{12}$, $\frac{4}{12}$

$\frac{1}{12}$, $\frac{1}{12}$, $\frac{3}{12}$

47 $\frac{1}{2}$, $\frac{5}{12}$, $\frac{7}{12}$

$\frac{6}{12}$, $\frac{5}{12}$, $\frac{7}{12}$

$\frac{5}{12}$, $\frac{1}{12}$, $\frac{7}{12}$

48 $\frac{1}{6}$, $\frac{3}{4}$, $\frac{1}{2}$

49 $\frac{1}{2}$, $\frac{2}{3}$, $\frac{1}{4}$

50 $\frac{11}{12}$, $\frac{1}{4}$, $\frac{5}{6}$

51 $\frac{1}{6}$, $\frac{1}{2}$, $\frac{5}{12}$

52 $\frac{5}{6}$, $\frac{1}{2}$, $\frac{1}{3}$

Easy

Questions 1–30

Add or subtract the tops.
Keep the bottom. Simplify.



WE DO With your teacher: copy the working.

E1 $\frac{4}{5} - \frac{1}{5}$ _____

E2 $\frac{2}{3} + \frac{2}{3}$ _____

YOU DO Your turn.

- 1 $\frac{1}{4} + \frac{3}{4}$ _____ **1**
- 2 $\frac{5}{10} + \frac{7}{10}$ _____
- 3 $\frac{6}{10} - \frac{4}{10}$ _____
- 4 $\frac{5}{10} + \frac{1}{10}$ _____
- 5 $\frac{6}{9} + \frac{3}{9}$ _____
- 6 $\frac{4}{5} + \frac{1}{5}$ _____
- 7 $\frac{5}{7} + \frac{1}{7}$ _____
- 8 $\frac{4}{6} + \frac{4}{6}$ _____
- 9 $\frac{6}{7} + \frac{3}{7}$ _____
- 10 $\frac{5}{7} + \frac{4}{7}$ _____
- 11 $\frac{7}{11} + \frac{8}{11}$ _____
- 12 $\frac{3}{4} - \frac{2}{4}$ _____
- 13 $\frac{5}{8} + \frac{6}{8}$ _____
- 14 $\frac{1}{3} + \frac{1}{3}$ _____
- 15 $\frac{4}{8} - \frac{3}{8}$ _____
- 16 $\frac{2}{3} - \frac{1}{3}$ _____
- 17 $\frac{2}{5} + \frac{3}{5}$ _____
- 18 $\frac{7}{11} + \frac{1}{11}$ _____
- 19 $\frac{6}{11} + \frac{6}{11}$ _____
- 20 $\frac{5}{11} + \frac{6}{11}$ _____
- 21 $\frac{2}{12} - \frac{1}{12}$ _____
- 22 $\frac{8}{11} - \frac{5}{11}$ _____
- 23 $\frac{10}{11} - \frac{9}{11}$ _____
- 24 $\frac{2}{10} + \frac{2}{10}$ _____
- 25 $\frac{4}{5} + \frac{4}{5}$ _____
- 26 $\frac{1}{10} + \frac{8}{10}$ _____
- 27 $\frac{1}{4} + \frac{2}{4}$ _____
- 28 $\frac{5}{8} - \frac{1}{8}$ _____
- 29 $\frac{3}{12} + \frac{5}{12}$ _____
- 30 $\frac{7}{8} + \frac{5}{8}$ _____

Medium

Questions 31–46

Use the lowest common denominator, then add or subtract.

WE DO With your teacher: copy the working.

E1 $\frac{1}{2} - \frac{1}{3}$ _____

E2 $\frac{2}{3} + \frac{1}{4}$ _____

YOU DO Your turn.

- 31 $\frac{2}{5} - \frac{1}{4}$
= $\frac{8}{20} - \frac{5}{20}$
= $\frac{3}{20}$
- 32 $\frac{2}{5} + \frac{2}{3}$
= $\frac{6}{15} + \frac{10}{15}$
= $\frac{16}{15}$
- 33 $\frac{2}{3} + \frac{7}{8}$
= $\frac{16}{24} + \frac{21}{24}$
= $\frac{37}{24}$
- 34 $\frac{2}{3} - \frac{1}{8}$ _____
- 35 $\frac{4}{5} + \frac{7}{8}$ _____
- 36 $\frac{1}{2} + \frac{1}{10}$ _____
- 37 $\frac{4}{5} + \frac{5}{6}$ _____
- 38 $\frac{1}{2} + \frac{3}{8}$ _____
- 39 $\frac{2}{3} + \frac{7}{10}$ _____
- 40 $\frac{2}{5} - \frac{1}{3}$ _____
- 41 $\frac{3}{5} + \frac{5}{8}$ _____
- 42 $\frac{2}{3} - \frac{1}{10}$ _____
- 43 $\frac{2}{3} - \frac{3}{8}$ _____
- 44 $\frac{3}{4} + \frac{2}{3}$ _____
- 45 $\frac{1}{2} - \frac{1}{10}$ _____
- 46 $\frac{3}{5} + \frac{7}{10}$ _____

Challenging

Questions 47–51

Three fractions: use one common denominator for all of them.



WE DO With your teacher: copy the working.

E1 $\frac{1}{2} + \frac{1}{6} - \frac{1}{12}$ _____

E2 $\frac{1}{2} + \frac{1}{3} - \frac{1}{6}$ _____

YOU DO Your turn.

- 47 $\frac{1}{3} + \frac{1}{4} - \frac{1}{6}$
= $\frac{4}{12} + \frac{3}{12} - \frac{2}{12}$
= $\frac{5}{12}$
- 48 $\frac{1}{4} + \frac{1}{3} - \frac{1}{12}$
= $\frac{3}{12} + \frac{4}{12} - \frac{1}{12}$
= $\frac{6}{12}$
= $\frac{1}{2}$
- 49 $\frac{1}{3} + \frac{1}{4} - \frac{1}{12}$ _____
- 50 $\frac{1}{3} + \frac{1}{6} - \frac{1}{12}$ _____
- 51 $\frac{1}{2} + \frac{1}{4} - \frac{1}{6}$ _____

4.04 Adding and subtracting mixed numerals

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Easy

Questions 1–30

Add the whole numbers, then add the fractions.



WE DO With your teacher: copy the working.

E1 $1\frac{1}{4} + 5\frac{1}{4}$

E2 $1\frac{2}{5} + 4\frac{2}{5}$

YOU DO Your turn.

1 $5\frac{1}{7} + 1\frac{3}{7}$

$6\frac{4}{7}$

2 $4\frac{1}{8} + 1\frac{3}{8}$

3 $6\frac{1}{3} + 3\frac{1}{3}$

4 $6\frac{1}{7} + 1\frac{1}{7}$

5 $3\frac{1}{4} + 2\frac{1}{4}$

6 $7\frac{3}{5} + 1\frac{1}{5}$

7 $7\frac{1}{5} + 4\frac{2}{5}$

8 $1\frac{1}{7} + 5\frac{4}{7}$

9 $2\frac{2}{5} + 3\frac{2}{5}$

10 $3\frac{1}{5} + 4\frac{3}{5}$

11 $4\frac{3}{8} + 1\frac{1}{8}$

12 $4\frac{1}{3} + 1\frac{1}{3}$

13 $2\frac{1}{3} + 2\frac{1}{3}$

14 $2\frac{1}{7} + 1\frac{5}{7}$

15 $6\frac{1}{7} + 2\frac{2}{7}$

16 $1\frac{1}{5} + 1\frac{1}{5}$

17 $3\frac{1}{9} + 5\frac{4}{9}$

18 $7\frac{2}{7} + 4\frac{4}{7}$

19 $6\frac{1}{4} + 5\frac{1}{4}$

20 $6\frac{4}{9} + 3\frac{1}{9}$

21 $7\frac{1}{3} + 5\frac{1}{3}$

22 $2\frac{5}{9} + 1\frac{2}{9}$

23 $3\frac{1}{7} + 2\frac{3}{7}$

24 $4\frac{7}{9} + 2\frac{1}{9}$

25 $5\frac{3}{5} + 5\frac{1}{5}$

26 $4\frac{3}{5} + 5\frac{1}{5}$

27 $4\frac{1}{3} + 3\frac{1}{3}$

28 $3\frac{1}{5} + 3\frac{2}{5}$

29 $7\frac{2}{5} + 1\frac{1}{5}$

30 $1\frac{1}{7} + 1\frac{4}{7}$

Medium

Questions 31–46

Change to improper fractions, then subtract.

WE DO With your teacher: copy the working.

E1 $5\frac{1}{3} - 3\frac{2}{3}$

E2 $6\frac{1}{5} - 1\frac{4}{5}$

YOU DO Your turn.

31 $3\frac{1}{3} - 1\frac{2}{3}$

$= \frac{10}{3} - \frac{5}{3}$

$= \frac{5}{3} = 1\frac{2}{3}$

32 $4\frac{1}{3} - 1\frac{2}{3}$

$= \frac{13}{3} - \frac{5}{3}$

$= \frac{8}{3} = \frac{\quad}{\quad} = \frac{2}{\quad}$

33 $7\frac{3}{5} - 1\frac{4}{5}$

$= \frac{38}{5} - \frac{9}{5}$

$= \frac{29}{5} = \frac{\quad}{\quad} = \frac{4}{\quad}$

34 $7\frac{1}{3} - 2\frac{2}{3}$

35 $6\frac{3}{5} - 4\frac{4}{5}$

36 $4\frac{1}{6} - 3\frac{5}{6}$

37 $4\frac{1}{8} - 1\frac{3}{8}$

38 $6\frac{3}{5} - 5\frac{4}{5}$

39 $5\frac{3}{5} - 2\frac{4}{5}$

40 $6\frac{1}{4} - 4\frac{3}{4}$

41 $7\frac{1}{5} - 5\frac{2}{5}$

42 $4\frac{5}{8} - 1\frac{7}{8}$

43 $4\frac{1}{3} - 2\frac{2}{3}$

44 $3\frac{1}{3} - 2\frac{2}{3}$

45 $4\frac{1}{4} - 1\frac{3}{4}$

46 $7\frac{1}{3} - 1\frac{2}{3}$

Challenging

Questions 47–51

Different denominators: change to improper fractions with a common denominator.



WE DO With your teacher: copy the working.

E1 $4\frac{1}{4} + 2\frac{4}{5}$

E2 $2\frac{3}{4} + 1\frac{1}{3}$

YOU DO Your turn.

47 $2\frac{2}{3} + 2\frac{2}{5}$

$= \frac{8}{3} + \frac{12}{5}$

$= \frac{40}{15} + \frac{36}{15}$

$= \frac{\quad}{\quad} = \frac{1}{\quad}$

48 $2\frac{2}{3} + 3\frac{5}{6}$

$= \frac{8}{3} + \frac{23}{6}$

$= \frac{16}{6} + \frac{23}{6}$

$= \frac{\quad}{\quad} = \frac{1}{\quad}$

49 $4\frac{2}{3} + 1\frac{3}{5}$

50 $4\frac{1}{2} + 2\frac{2}{3}$

51 $4\frac{2}{3} + 3\frac{3}{5}$

4.05 Fraction of a quantity

Easy

Questions 1–29

Find the fraction of the amount: divide by the denominator.



WE DO With your teacher: copy the working.

E1 $\frac{1}{8}$ of 80 _____

E2 $\frac{1}{10}$ of 70 _____

YOU DO Your turn.

1 $\frac{1}{10}$ of 90 _____ **9**

2 $\frac{1}{10}$ of 80 _____

3 $\frac{1}{3}$ of 9 _____

4 $\frac{1}{2}$ of 8 _____

5 $\frac{1}{4}$ of 28 _____

6 $\frac{1}{12}$ of 24 _____

7 $\frac{1}{12}$ of 36 _____

8 $\frac{1}{7}$ of 84 _____

9 $\frac{1}{9}$ of 18 _____

10 $\frac{1}{5}$ of 35 _____

11 $\frac{1}{4}$ of 8 _____

12 $\frac{1}{6}$ of 72 _____

13 $\frac{1}{2}$ of 6 _____

14 $\frac{1}{4}$ of 48 _____

15 $\frac{1}{5}$ of 50 _____

16 $\frac{1}{5}$ of 30 _____

17 $\frac{1}{4}$ of 24 _____

18 $\frac{1}{4}$ of 16 _____

19 $\frac{1}{2}$ of 16 _____

20 $\frac{1}{12}$ of 72 _____

21 $\frac{1}{8}$ of 56 _____

22 $\frac{1}{5}$ of 55 _____

23 $\frac{1}{10}$ of 40 _____

24 $\frac{1}{5}$ of 40 _____

25 $\frac{1}{12}$ of 60 _____

26 $\frac{1}{4}$ of 32 _____

27 $\frac{1}{4}$ of 40 _____

28 $\frac{1}{3}$ of 36 _____

29 $\frac{1}{2}$ of 24 _____

Medium

Questions 30–49

Divide by the denominator, then multiply by the numerator.

WE DO With your teacher: copy the working.

E1 $\frac{3}{4}$ of 28 _____

E2 $\frac{2}{3}$ of 21 _____

YOU DO Your turn.

30 $\frac{8}{9}$ of 90 _____

31 $\frac{4}{5}$ of 15 _____

$90 \div 9 = 10$

$15 \div 5 = 3$

$10 \times 8 = 80$

$___ \times ___ = ___$

32 $\frac{3}{5}$ of 35 _____

33 $\frac{2}{3}$ of 18 _____

$___ \div ___ = ___$

$___ \times ___ = ___$

34 $\frac{2}{3}$ of 9 _____

35 $\frac{4}{5}$ of 40 _____

36 $\frac{5}{8}$ of 88 _____

37 $\frac{3}{7}$ of 70 _____

38 $\frac{3}{8}$ of 24 _____

39 $\frac{3}{4}$ of 32 _____

40 $\frac{3}{4}$ of 16 _____

41 $\frac{2}{5}$ of 45 _____

42 $\frac{4}{5}$ of 20 _____

43 $\frac{3}{10}$ of 90 _____

44 $\frac{8}{9}$ of 27 _____

45 $\frac{5}{6}$ of 54 _____

46 $\frac{3}{10}$ of 40 _____

47 $\frac{4}{5}$ of 50 _____

48 $\frac{2}{9}$ of 45 _____

49 $\frac{2}{3}$ of 36 _____

Challenging

Questions 50–55

Change to the smaller unit first, then find the fraction.



WE DO With your teacher: copy the working.

E1 $\frac{1}{3}$ of a day, in hours _____

E2 $\frac{3}{5}$ of an hour, in minutes _____

YOU DO Your turn.

50 $\frac{1}{12}$ of an hour, in minutes _____

an hour = 60 minutes

$60 \div 12 = 5$

$___ \times ___ = ___ \text{ minutes}$

51 $\frac{5}{12}$ of an hour, in minutes _____

an hour = $___ \text{ minutes}$

$___ \div ___ = ___$

$___ \times ___ = ___ \text{ minutes}$

52 $\frac{2}{5}$ of an hour, in minutes _____

53 $\frac{1}{4}$ of \$1, in cents _____

54 $\frac{3}{5}$ of 1 kg, in grams _____

55 $\frac{1}{10}$ of \$1, in cents _____

Easy

Questions 1–30

Multiply the tops, multiply the bottoms. Simplify.



WE DO With your teacher: copy the working.

E1 $\frac{8}{9} \times \frac{1}{2}$

E2 $\frac{1}{2} \times \frac{4}{5}$

YOU DO Your turn.

1 $\frac{3}{6} \times \frac{1}{3}$

$\frac{1}{6}$

2 $\frac{2}{8} \times \frac{2}{3}$

3 $\frac{4}{5} \times \frac{5}{7}$

4 $\frac{2}{5} \times \frac{2}{4}$

5 $\frac{1}{2} \times \frac{1}{2}$

6 $\frac{1}{2} \times \frac{3}{8}$

7 $\frac{1}{2} \times \frac{1}{4}$

8 $\frac{4}{5} \times \frac{6}{9}$

9 $\frac{1}{6} \times \frac{2}{5}$

10 $\frac{3}{4} \times \frac{1}{6}$

11 $\frac{2}{3} \times \frac{1}{7}$

12 $\frac{2}{5} \times \frac{3}{9}$

13 $\frac{1}{3} \times \frac{3}{5}$

14 $\frac{5}{9} \times \frac{3}{6}$

15 $\frac{1}{8} \times \frac{4}{6}$

16 $\frac{2}{5} \times \frac{2}{5}$

17 $\frac{2}{4} \times \frac{6}{9}$

18 $\frac{1}{6} \times \frac{1}{5}$

19 $\frac{3}{5} \times \frac{5}{7}$

20 $\frac{4}{6} \times \frac{2}{9}$

21 $\frac{4}{8} \times \frac{2}{6}$

22 $\frac{3}{6} \times \frac{2}{5}$

23 $\frac{1}{5} \times \frac{5}{9}$

24 $\frac{3}{6} \times \frac{1}{2}$

25 $\frac{1}{2} \times \frac{3}{6}$

26 $\frac{2}{3} \times \frac{2}{4}$

27 $\frac{2}{4} \times \frac{3}{4}$

28 $\frac{3}{4} \times \frac{1}{4}$

29 $\frac{3}{9} \times \frac{6}{9}$

30 $\frac{7}{8} \times \frac{1}{4}$

Medium

Questions 31–46

Cancel first (divide a top and a bottom by the same number), then multiply.

WE DO With your teacher: copy the working.

E1 $\frac{2}{5} \times \frac{1}{14}$

E2 $\frac{4}{5} \times \frac{3}{14}$

YOU DO Your turn.

31 $\frac{4}{7} \times \frac{3}{14}$

$= \frac{2}{7} \times \frac{3}{7}$

$= \frac{6}{49}$

32 $\frac{2}{5} \times \frac{2}{4}$

$= \frac{1}{5} \times \frac{2}{2}$

$= \frac{1}{5}$

33 $\frac{6}{7} \times \frac{1}{10}$

$= \frac{3}{7} \times \frac{1}{5}$

$= \frac{3}{35}$

34 $\frac{6}{7} \times \frac{2}{14}$

35 $\frac{4}{5} \times \frac{2}{10}$

36 $\frac{2}{7} \times \frac{2}{10}$

37 $\frac{3}{5} \times \frac{1}{6}$

38 $\frac{2}{3} \times \frac{2}{12}$

39 $\frac{6}{7} \times \frac{3}{10}$

40 $\frac{4}{5} \times \frac{1}{10}$

41 $\frac{4}{7} \times \frac{3}{16}$

42 $\frac{2}{5} \times \frac{2}{10}$

43 $\frac{2}{4} \times \frac{1}{4}$

44 $\frac{6}{7} \times \frac{4}{8}$

45 $\frac{3}{4} \times \frac{3}{9}$

46 $\frac{6}{7} \times \frac{1}{21}$

Challenging

Questions 47–51

Change to improper fractions, cancel, then multiply.

WE DO With your teacher: copy the working.

E1 $1\frac{1}{3} \times 2\frac{3}{5}$

E2 $1\frac{3}{4} \times 1\frac{2}{3}$

YOU DO Your turn.

47 $2\frac{4}{5} \times 1\frac{1}{2}$

$= \frac{14}{5} \times \frac{3}{2}$

$= \frac{42}{10}$

$= \frac{21}{5}$

48 $1\frac{3}{4} \times 3\frac{4}{5}$

$= \frac{7}{4} \times \frac{19}{5}$

$= \frac{133}{20}$

$= \frac{13}{2}$

49 $1\frac{1}{3} \times 1\frac{1}{3}$

50 $3\frac{4}{5} \times 1\frac{1}{2}$

51 $2\frac{1}{5} \times 3\frac{2}{4}$

4.07 Dividing fractions

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Easy

Questions 1–30

Write the reciprocal: turn the fraction upside down.



WE DO With your teacher: copy the working.

E1 $\frac{4}{8}$

E2 $\frac{1}{7}$

YOU DO Your turn.

1 $\frac{1}{2}$

2

2 $\frac{11}{12}$

3 $\frac{9}{4}$

4 $\frac{5}{2}$

5 $\frac{3}{9}$

6 $\frac{8}{10}$

7 $\frac{5}{12}$

8 $\frac{11}{4}$

9 $\frac{4}{11}$

10 $\frac{7}{8}$

11 $\frac{11}{5}$

12 $\frac{1}{5}$

13 $\frac{3}{7}$

14 $\frac{5}{10}$

15 $\frac{6}{11}$

16 $\frac{4}{5}$

17 $\frac{10}{3}$

18 $\frac{7}{12}$

19 $\frac{5}{7}$

20 $\frac{9}{5}$

21 $\frac{8}{11}$

22 $\frac{6}{12}$

23 $\frac{6}{7}$

24 $\frac{6}{9}$

25 $\frac{2}{10}$

26 $\frac{5}{4}$

27 $\frac{2}{8}$

28 $\frac{6}{2}$

29 $\frac{6}{4}$

30 $\frac{10}{5}$

Medium

Questions 31–46

Keep, change, flip: multiply by the reciprocal of the second fraction.

WE DO With your teacher: copy the working.

E1 $\frac{5}{7} \div \frac{4}{5}$

E2 $\frac{4}{9} \div \frac{6}{7}$

YOU DO Your turn.

31 $\frac{1}{9} \div \frac{3}{6}$

$= \frac{1}{9} \times \frac{6}{3}$

$= \frac{2}{9}$

32 $\frac{4}{9} \div \frac{1}{4}$

$= \frac{4}{9} \times \frac{4}{1}$

$= \frac{16}{9}$

33 $\frac{5}{8} \div \frac{3}{6}$

$= \frac{5}{8} \times \frac{6}{3}$

$= \frac{5}{4}$

34 $\frac{2}{7} \div \frac{1}{3}$

35 $\frac{6}{9} \div \frac{3}{4}$

36 $\frac{5}{7} \div \frac{1}{3}$

37 $\frac{2}{6} \div \frac{4}{5}$

38 $\frac{2}{9} \div \frac{4}{5}$

39 $\frac{1}{2} \div \frac{2}{8}$

40 $\frac{2}{8} \div \frac{3}{6}$

41 $\frac{5}{7} \div \frac{2}{4}$

42 $\frac{1}{9} \div \frac{1}{2}$

43 $\frac{1}{2} \div \frac{3}{8}$

44 $\frac{3}{5} \div \frac{2}{3}$

45 $\frac{4}{6} \div \frac{1}{3}$

46 $\frac{2}{4} \div \frac{1}{4}$

Challenging

Questions 47–51

Change to improper fractions, then keep, change, flip.



WE DO With your teacher: copy the working.

E1 $2\frac{1}{5} \div 2\frac{1}{3}$

E2 $3\frac{1}{4} \div 1\frac{3}{5}$

YOU DO Your turn.

47 $2\frac{1}{2} \div 2\frac{1}{2}$

$= \frac{5}{2} \div \frac{5}{2}$

$= \frac{5}{2} \times \frac{2}{5}$

$= 1$

48 $3\frac{1}{3} \div 1\frac{1}{4}$

$= \frac{10}{3} \div \frac{5}{4}$

$= \frac{10}{3} \times \frac{4}{5}$

$= \frac{40}{3}$

49 $4\frac{1}{2} \div 1\frac{1}{5}$

50 $2\frac{2}{5} \div 2\frac{2}{3}$

51 $4\frac{1}{5} \div 1\frac{1}{5}$

Easy

Questions 1–21

What fraction is left? The whole is 1.



WE DO With your teacher: copy the working.

E1 $\frac{2}{5}$ of a cake is sold _____

E2 $\frac{2}{3}$ of a cake is sold _____

YOU DO Your turn.

1 $\frac{2}{3}$ of a tank of petrol is used $\frac{1}{3}$

2 $\frac{11}{12}$ of a class is away _____

3 $\frac{11}{12}$ of a bottle of water is drunk _____

4 $\frac{3}{4}$ of a bottle of water is drunk _____

5 $\frac{2}{5}$ of a bottle of water is drunk _____

6 $\frac{1}{4}$ of a pizza is eaten _____

7 $\frac{1}{2}$ of a book is read _____

8 $\frac{1}{10}$ of a book is read _____

9 $\frac{1}{6}$ of a bottle of water is drunk _____

10 $\frac{5}{6}$ of a class is away _____

11 $\frac{7}{8}$ of a book is read _____

12 $\frac{1}{3}$ of a tank of petrol is used _____

13 $\frac{1}{8}$ of a tank of petrol is used _____

14 $\frac{1}{3}$ of a class is away _____

15 $\frac{1}{2}$ of a class is away _____

16 $\frac{1}{3}$ of a book is read _____

17 $\frac{3}{5}$ of a book is read _____

18 $\frac{1}{2}$ of a tank of petrol is used _____

19 $\frac{1}{2}$ of a pizza is eaten _____

20 $\frac{2}{5}$ of a tank of petrol is used _____

21 $\frac{3}{5}$ of a class is away _____

Medium

Questions 22–28

Find the fraction of the amount, then answer the question.

WE DO With your teacher: copy the working.

E1 Sam had \$120 and spent $\frac{3}{10}$ of it. How much is left?

E2 Sam had \$45 and spent $\frac{4}{5}$ of it. How much is left?

YOU DO Your turn.

22 Sam had \$70 and spent $\frac{3}{10}$ of it. How much is left?

spent: $70 \div 10 \times 3 = \$21$

left: $70 - 21 = \$49$

23 Sam had \$24 and spent $\frac{1}{3}$ of it. How much is left?

spent: $24 \div 3 \times 1 = \$8$

left: _____ - _____ = _____

24 Sam had \$18 and spent $\frac{1}{3}$ of it. How much is left?

spent: _____ $\div$ _____ $\times$ _____ = _____

left: _____ - _____ = _____

25 Sam had \$44 and spent $\frac{1}{4}$ of it. How much is left?

26 Sam had \$96 and spent $\frac{1}{8}$ of it. How much is left?

27 Sam had \$24 and spent $\frac{1}{4}$ of it. How much is left?

28 Sam had \$72 and spent $\frac{5}{8}$ of it. How much is left?

Challenging

Questions 29–32

Two fractions of the same whole. Find each part, then the rest.

WE DO With your teacher: copy the working.

E1 12 students: $\frac{1}{3}$ walk to school and $\frac{1}{6}$ ride. The rest take the bus. How many take the bus?

E2 80 students: $\frac{1}{4}$ walk to school and $\frac{1}{5}$ ride. The rest take the bus. How many take the bus?

YOU DO Your turn.

29 48 students: $\frac{1}{3}$ walk to school and $\frac{1}{4}$ ride. The rest take the bus. How many take the bus?

walk: $48 \div 3 = 16$

ride: $48 \div 4 = 12$

bus: _____ - _____ - _____ = _____

30 12 students: $\frac{1}{2}$ walk to school and $\frac{1}{4}$ ride. The rest take the bus. How many take the bus?

walk: _____ $\div$ _____ = _____

ride: _____ $\div$ _____ = _____

bus: _____ - _____ - _____ = _____

31 30 students: $\frac{1}{3}$ walk to school and $\frac{1}{5}$ ride. The rest take the bus. How many take the bus?

32 50 students: $\frac{1}{2}$ walk to school and $\frac{1}{5}$ ride. The rest take the bus. How many take the bus?

4.09 Fractions on a calculator

Easy

Questions 1–30

Use the fraction key on your calculator.



WE DO With your teacher: copy the working.

E1 $\frac{2}{10} + \frac{9}{19}$ _____

E2 $\frac{4}{14} + \frac{9}{12}$ _____

YOU DO Your turn.

1 $\frac{5}{9} \times \frac{4}{19}$ 20
171

2 $\frac{6}{15} + \frac{9}{10}$ _____

3 $\frac{5}{10} \times \frac{4}{16}$ _____

4 $\frac{3}{20} \times \frac{3}{6}$ _____

5 $\frac{14}{19} \times \frac{9}{16}$ _____

6 $\frac{6}{15} \times \frac{13}{16}$ _____

7 $\frac{9}{19} + \frac{6}{10}$ _____

8 $\frac{2}{5} + \frac{8}{16}$ _____

9 $\frac{6}{13} + \frac{5}{16}$ _____

10 $\frac{8}{10} \times \frac{10}{12}$ _____

11 $\frac{1}{14} + \frac{11}{16}$ _____

12 $\frac{7}{10} + \frac{5}{7}$ _____

13 $\frac{4}{6} \times \frac{5}{11}$ _____

14 $\frac{6}{12} \times \frac{3}{6}$ _____

15 $\frac{4}{14} + \frac{12}{13}$ _____

16 $\frac{3}{5} \times \frac{1}{13}$ _____

17 $\frac{5}{9} + \frac{3}{6}$ _____

18 $\frac{6}{18} + \frac{1}{5}$ _____

19 $\frac{5}{8} + \frac{5}{18}$ _____

20 $\frac{1}{16} + \frac{1}{17}$ _____

21 $\frac{1}{12} + \frac{14}{16}$ _____

22 $\frac{18}{19} + \frac{4}{6}$ _____

23 $\frac{4}{5} + \frac{9}{13}$ _____

24 $\frac{14}{17} \times \frac{6}{8}$ _____

25 $\frac{8}{12} + \frac{3}{16}$ _____

26 $\frac{9}{11} + \frac{3}{8}$ _____

27 $\frac{15}{18} \times \frac{2}{6}$ _____

28 $\frac{10}{15} + \frac{1}{5}$ _____

29 $\frac{4}{12} + \frac{6}{7}$ _____

30 $\frac{5}{9} \times \frac{10}{17}$ _____

Medium

Questions 31–38

Use your calculator. Write what you typed, then the answer.

WE DO With your teacher: copy the working.

E1 $\frac{11}{14} \div \frac{3}{6}$ _____

E2 $\frac{10}{16} \div \frac{11}{13}$ _____

YOU DO Your turn.

31 $\frac{11}{19} \div \frac{2}{11}$ _____

typed: $11/19 \div 2/11$

= $3 \frac{7}{38}$

32 $\frac{9}{14} \div \frac{6}{7}$ _____

typed: $9/14 \div 6/7$

= $\frac{3}{4}$

33 $\frac{3}{8} \div \frac{2}{6}$ _____

typed: ____/____ ÷ ____/____

= $\frac{1}{2}$

34 $\frac{6}{8} \div \frac{4}{8}$ _____

35 $\frac{8}{10} \div \frac{3}{5}$ _____

36 $\frac{2}{10} \div \frac{8}{9}$ _____

37 $\frac{6}{8} \div \frac{5}{19}$ _____

38 $\frac{11}{20} \div \frac{9}{12}$ _____

Challenging

Questions 39–45

Use your calculator and brackets. Check the answer is sensible.



WE DO With your teacher: copy the working.

E1 $(\frac{1}{4} + \frac{3}{7}) \times \frac{6}{9}$ _____

E2 $(\frac{2}{8} + \frac{1}{7}) \times \frac{2}{3}$ _____

YOU DO Your turn.

39 $(\frac{4}{5} + \frac{5}{6}) \times \frac{1}{5}$ _____

bracket: $1 \frac{19}{30}$

$\times \frac{1}{5} = \frac{49}{30}$

40 $(\frac{8}{9} + \frac{2}{3}) \times \frac{5}{6}$ _____

bracket: ____ $\frac{5}{6}$

$\times \frac{5}{6} = \frac{8}{3}$

41 $(\frac{3}{8} + \frac{6}{9}) \times \frac{3}{8}$ _____

42 $(\frac{1}{2} + \frac{2}{3}) \times \frac{1}{9}$ _____

43 $(\frac{4}{6} + \frac{2}{8}) \times \frac{8}{9}$ _____

44 $(\frac{7}{8} + \frac{6}{9}) \times \frac{3}{9}$ _____

45 $(\frac{7}{8} + \frac{1}{4}) \times \frac{1}{9}$ _____

Easy

Questions 1–25

Write each percentage as a fraction out of 100, then simplify.



WE DO With your teacher: copy the working.

E1 40%

E2 45%

YOU DO Your turn.

1 8%

$\frac{2}{25}$

2 35%

3 10%

4 4%

5 60%

6 80%

7 2%

8 75%

9 15%

10 30%

11 70%

12 20%

Write each percentage as a decimal: divide by 100.



WE DO With your teacher: copy the working.

E3 96%

YOU DO Your turn.

13 38%

0.38

14 16%

15 147%

16 90%

17 89%

18 66%

19 53%

20 146%

21 2%

22 130%

23 64%

24 84%

25 149%

Medium

Questions 26–43

Write the fraction out of 100, then as a percentage.

WE DO

With your teacher: copy the working.

E1 $\frac{3}{4}$

E2 $\frac{3}{10}$

YOU DO Your turn.

26 $\frac{9}{10}$

27 $\frac{1}{5}$

= $\frac{90}{100}$

= $\frac{20}{100}$

= 90%

= ___%

28 $\frac{1}{4}$

29 $\frac{21}{25}$

= $\frac{25}{100}$

= ___%

30 $\frac{2}{5}$

31 $\frac{13}{20}$

32 $\frac{4}{25}$

33 $\frac{1}{2}$

34 $\frac{11}{25}$

35 $\frac{14}{25}$

36 $\frac{2}{25}$

37 $\frac{3}{50}$

38 $\frac{8}{25}$

39 $\frac{3}{5}$

40 $\frac{22}{25}$

41 $\frac{6}{25}$

42 $\frac{23}{25}$

43 $\frac{16}{25}$

Challenging

Questions 44–51

Write each decimal as a percentage and as a fraction in simplest form.



WE DO With your teacher: copy the working.

E1 0.6

E2 0.75

YOU DO Your turn.

44 0.1

× 100: 10%

$\frac{10}{100} = \frac{1}{10}$

45 0.25

× ___: ___%

$\frac{25}{100} = \frac{1}{4}$

46 0.04

47 0.3

48 0.12

49 0.7

50 0.2

51 0.35

4.11 Percentage of a quantity

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Easy

Questions 1–30

Find mentally. 10% is $\div 10$, 50% is half, 25% is a quarter.



WE DO With your teacher: copy the working.

E1 50% of 142 _____

E2 25% of 32 _____

YOU DO Your turn.

1 1% of 900 _____ **9**

2 1% of 500 _____

3 50% of 170 _____

4 25% of 64 _____

5 1% of 800 _____

6 1% of 300 _____

7 50% of 50 _____

8 10% of 660 _____

9 25% of 148 _____

10 10% of 520 _____

11 50% of 16 _____

12 10% of 160 _____

13 10% of 290 _____

14 25% of 124 _____

15 1% of 400 _____

16 10% of 720 _____

17 1% of 200 _____

18 10% of 570 _____

19 10% of 730 _____

20 25% of 88 _____

21 50% of 44 _____

22 10% of 810 _____

23 25% of 60 _____

24 25% of 36 _____

25 10% of 30 _____

26 25% of 40 _____

27 25% of 44 _____

28 50% of 106 _____

29 10% of 490 _____

30 10% of 670 _____

Medium

Questions 31–52

Find 10% first, then build the percentage you need.

WE DO With your teacher: copy the working.

E1 15% of 160 _____

E2 70% of 280 _____

YOU DO Your turn.

31 5% of 300 _____

32 70% of 260 _____

10% = 30

10% = 26

5% = 30 $\div$ 2 = 15

___% = ___ $\times$ ___

33 60% of 340 _____

34 60% of 480 _____

___% = ___

___% = ___ $\times$ ___

35 30% of 540 _____

36 30% of 480 _____

37 15% of 200 _____

38 60% of 380 _____

39 30% of 60 _____

40 20% of 480 _____

41 15% of 600 _____

42 20% of 100 _____

43 60% of 160 _____

44 40% of 580 _____

45 40% of 500 _____

46 30% of 600 _____

47 20% of 440 _____

48 60% of 100 _____

49 60% of 260 _____

50 20% of 140 _____

51 5% of 320 _____

52 20% of 120 _____

Challenging

Questions 53–60

Find the discount, then the sale price.



WE DO With your teacher: copy the working.

E1 A \$200 jacket is 15% off. Find the sale price.

E2 A \$40 jacket is 40% off. Find the sale price.

YOU DO Your turn.

53 A \$60 jacket is 50% off. Find the sale price.

discount: 50% of 60 = \$30

sale price: ___ - ___ = ___

54 A \$260 jacket is 40% off. Find the sale price.

discount: ___% of ___ = ___

sale price: ___ - ___ = ___

55 A \$90 jacket is 30% off. Find the sale price.

56 A \$160 jacket is 40% off. Find the sale price.

57 A \$140 jacket is 30% off. Find the sale price.

58 A \$210 jacket is 15% off. Find the sale price.

59 A \$160 jacket is 10% off. Find the sale price.

60 A \$260 jacket is 50% off. Find the sale price.

4.12 Expressing as a fraction or percentage

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Easy

Questions 1–30

Write as a fraction, then as a percentage.



WE DO With your teacher: copy the working.

E1 2 out of 5

E2 3 out of 5

YOU DO Your turn.

1 11 out of 25

44%

2 4 out of 20

3 9 out of 10

4 4 out of 10

5 16 out of 50

6 1 out of 4

7 14 out of 20

8 1 out of 10

9 2 out of 20

10 2 out of 4

11 2 out of 10

12 3 out of 4

13 26 out of 50

14 17 out of 20

15 5 out of 10

16 19 out of 25

17 13 out of 50

18 9 out of 25

19 5 out of 50

20 9 out of 20

21 1 out of 20

22 9 out of 50

23 6 out of 20

24 25 out of 50

25 40 out of 50

26 6 out of 10

27 7 out of 10

28 11 out of 20

29 19 out of 20

30 2 out of 25

Medium

Questions 31–50

Write as a fraction, make the denominator 100, then write the percentage.



WE DO With your teacher: copy the working.

E1 8 out of 50

E2 14 out of 20

YOU DO Your turn.

31 1 out of 4

32 3 out of 4

$$\frac{1}{4} = \frac{25}{100}$$

$$= 25\%$$

$$\frac{3}{4} = \frac{75}{100}$$

$$= 75\%$$

33 28 out of 50

34 5 out of 25

$$\frac{28}{50} = \frac{56}{100}$$

$$= 56\%$$

35 2 out of 4

36 5 out of 10

37 14 out of 25

38 3 out of 5

39 13 out of 20

40 3 out of 50

41 2 out of 25

42 6 out of 20

43 1 out of 5

44 19 out of 20

45 1 out of 10

46 48 out of 50

47 6 out of 25

48 3 out of 20

49 17 out of 20

50 10 out of 25

Challenging

Questions 51–56

Change to the same unit first, then write as a percentage.



WE DO With your teacher: copy the working.

E1 250 g out of 2 kg

E2 75c out of \$3

YOU DO Your turn.

51 25c out of \$2

$$\$2 = 200c$$

$$\frac{25}{200} \times 100\%$$

$$= 12.5\%$$

52 25c out of \$1

$$25c = 0.25$$

$$\frac{0.25}{1} \times 100\%$$

$$= 25\%$$

53 15 min out of 1 h

54 45 min out of 2 h

55 20 cm out of 1 m

56 25 cm out of 2 m